A lower bound of 9/8 for the Schrijver–Seymour–Winkler ring loading constant

Maksim Olenev

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Abstract

Let β be the smallest number with the following property: for all m and all non-negative reals u_i, v_i $(1 \le i \le m)$ with $u_i + v_i \le 1$ one can choose $z_i \in \{v_i, -u_i\}$ with $\left|\sum_{i \le k} z_i - \sum_{i > k} z_i\right| \le \beta$ for every $1 \le k \le m$. Schrijver, Seymour and Winkler showed that every split routing in the Ring Loading Problem can be made unsplittable while increasing each edge load by at most βD, where D is the maximum demand. Previously, the best lower bound was $\beta \ge 47/42$, based on an instance found by AlphaEvolve. We give an explicit instance with $m = 14$ and all entries in $\frac{1}{16}\mathbb{Z}$ showing $\beta \ge 9/8$. The proof is a finite exact check over 2^{14} sign patterns.

1 The constant

Definition 1 (Schrijver, Seymour, Winkler [1])**.** For $u, v \in \mathbb{R}^m_{\ge 0}$ with $u_i + v_i \le 1$ let

$$\alpha(u, v) = \min_{z \in \prod_i \{v_i, -u_i\}} \max_{1 \le k \le m} \left| \sum_{i=1}^{k} z_i - \sum_{i=k+1}^{m} z_i \right|,$$

and let $\beta = \sup_m \sup_{u,v} \alpha(u, v)$.

Since $\alpha(u, v)$ is a minimum over a finite set, β is also the smallest number with the property stated in the abstract. Schrijver, Seymour and Winkler [1] introduced β in their analysis of the Ring Loading Problem: every split routing can be turned into an unsplittable routing while increasing the load of every edge by at most βD, where D is the maximum demand; the numbers u_i and v_i describe how split demands are divided between the two directions (see [1, 2] for the reduction). In the AlphaEvolve problem repository [5], this is Problem 61, where the constant is denoted C (Section 6.41 of [4]). Table 1 lists the known bounds.

Bound	Source	Year
$\beta \le 3/2, \quad \beta \ge 101/100$	Schrijver, Seymour, Winkler [1]	1998
$\beta \le 19/14, \quad \beta \ge 11/10$	Skutella [2]	2016
$\beta \le 13/10$	Däubel [3]	2022
$\beta \ge 1.1190475\ldots$	AlphaEvolve [4, 5]	2025
$\beta \ge 47/42 \approx 1.1190476$	Yadav [6] (exact certificate)	2026
$\beta \ge 9/8 = 1.125$	this note	2026

Table 1: Known bounds on β.

2 The certificate

Theorem 1. *For the instance in Table 2, $\alpha(u, v) = 9/8$. Consequently, $\beta \geq 9/8$.*

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$16\,u_i$	6	3	12	8	2	11	4	5	12	8	2	13	2	12
$16\,v_i$	2	13	2	8	12	5	4	11	2	8	12	3	6	4
$16(u_i + v_i)$	8	16	14	16	14	16	8	16	14	16	14	16	8	16

Table 2: An instance with $\alpha(u, v) = 9/8$ (all values are multiples of $1/16$).

Proof. All u_i, v_i are non-negative and $u_i + v_i \leq 1$. For a choice z write $p_k = \sum_{i \leq k} z_i$ and $y = p_m$; then $\sum_{i \leq k} z_i - \sum_{i > k} z_i = 2p_k - y$. Since every $16z_i$ is an integer, the quantities $16(2p_k - y)$ are integers, and $\alpha(u, v)$ can be computed exactly by enumerating the $2^{14} = 16384$ choices of z. This enumeration shows $\max_{1 \leq k \leq 14} |2p_k - y| \geq 18/16 = 9/8$ for every z, with equality for exactly 52 choices. Hence $\alpha(u, v) = 9/8$ and $\beta \geq \alpha(u, v) = 9/8$. □

Remark 1. The verification code in the AlphaEvolve notebook [5] uses $1 \leq k \leq m - 1$ instead of $1 \leq k \leq m$. Under that convention the instance of Table 2 has value only $3/4$. Adding a pair $(0, 0)$ at one end (or at both ends) makes the two conventions agree: the instance padded at both ends ($m = 16$) has value $9/8$ under either convention. In general, the value under the $1 \leq k \leq m - 1$ convention never exceeds $\alpha(u, v)$, and padding shows that both conventions define the same constant β.

3 Verification

The value was confirmed independently by (i) exhaustive enumeration in exact rational arithmetic (Python `fractions`, listed below); (ii) exhaustive enumeration in integer arithmetic, in C; (iii) an exact dynamic program over the prefix sums p_k for integer instances, in C; and (iv) the function `compute_alpha_exact` copied verbatim from the AlphaEvolve notebook [5], applied to the padded instance, which returns 1.125 (floating-point evaluation is exact here, since all entries are multiples of $1/16$). The code for (ii)–(iv) and for the search described below is distributed with the certificate files.

```
import itertools
from fractions import Fraction as F
P = [(6,2),(3,13),(12,2),(8,8),(2,12),(11,5),(4,4),
     (5,11),(12,2),(8,8),(2,12),(13,3),(2,6),(12,4)]
u = [F(a,16) for a,b in P]; v = [F(b,16) for a,b in P]; m = len(P)
vals = [max(abs(sum(z[:k]) - sum(z[k:])) for k in range(1, m+1))
        for z in itertools.product(*[(v[i], -u[i]) for i in range(m)])]
alpha = min(vals)
print(alpha, vals.count(alpha)) # 9/8 52
```

4 How the instance was found

The search had two stages. First, we ran simulated annealing over instances with entries in $\frac{1}{84}\mathbb{Z}$, restricted to palindromic instances ($u_{m+1-i} = v_i$) in which every second pair is full ($u_i + v_i = 1$). Each candidate was evaluated exactly by the dynamic program in (iii); ties were broken by the number of partial states of the dynamic program from which the current value can still be avoided, which turns plateaus into slopes. For $m \in \{17, 19, 21\}$ the best runs reached $94/84 = 47/42$, matching the previous record.

Second, the best outputs of the annealing (with values up to 94/84) were refined in continuous variables by sequential linear programming: the near-optimal sign patterns are linearized at their active index k, and an LP with a trust region is solved repeatedly. Several of these instances rose to exactly 9/8; others stopped at 31/28, 10/9, 29/26 or 47/42. The value of an instance with entries in $\frac{1}{84}\mathbb{Z}$ is a multiple of 1/84, and $9/8 = 94.5/84$ is not, so the grid search could not have found an instance with this value. Rationalization of the LP solution and deletion of redundant pairs gave the instance in Table 2.

5 Further observations

- The pair sums have period six: $(8, 16, 14, 16, 14, 16)$ twice, followed by $(8, 16)$, in units of 1/16; every second pair is full ($u_i + v_i = 1$).
- The first 13 pairs form a palindromic instance ($u_{14-i} = v_i$ for $1 \le i \le 13$), so the certificate is a palindromic 13-pair instance with one extra pair appended.
- A trust-region mixed-integer program (HiGHS, default tolerances) found no instance within ℓ_∞-distance 0.02 of the 14-pair certificate with value above 9/8. This is numerical evidence of local optimality, not a proof.
- Deleting any single pair from the certificate lowers the value to at most 1; the best value we found by re-optimizing the resulting 13-pair instances was about 1.094. We do not claim that $m = 14$ is the smallest size achieving 9/8.
- Skutella [2] conjectured that $L \le L^* + \frac{11}{10}D$ for every instance of the Ring Loading Problem, where L and L^* are the optimal unsplittable and split loads, and gave an explicit instance with $L = L^* + \frac{11}{10}D$. A lower bound on β does not by itself contradict this conjecture: β bounds the increase of every edge load for every split routing, whereas a counterexample would require an instance in which every unsplittable routing has maximum load greater than $L^* + \frac{11}{10}D$. Whether the present instance can be completed to such an example is left open.

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References

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[5] Google DeepMind. *AlphaEvolve repository of problems*, Problem 61 (The Ring Loading Problem), `https://github.com/google-deepmind/alphaevolve_repository_of_problems`. Accessed October 2026.

[6] V. Yadav. Exact certificate for the lower bound 47/42, issue #11 in [5], October 2026, `https://github.com/google-deepmind/alphaevolve_repository_of_problems/issues/11`.